MILITARY SPECIFICATION SHEET

CABLE ASSEMBLY, AIRCRAFT AUDIO, CONNECTOR (U-75/U)

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connector described herein shall consist of this document and the latest issue of specification MIL-C-22442(AS).

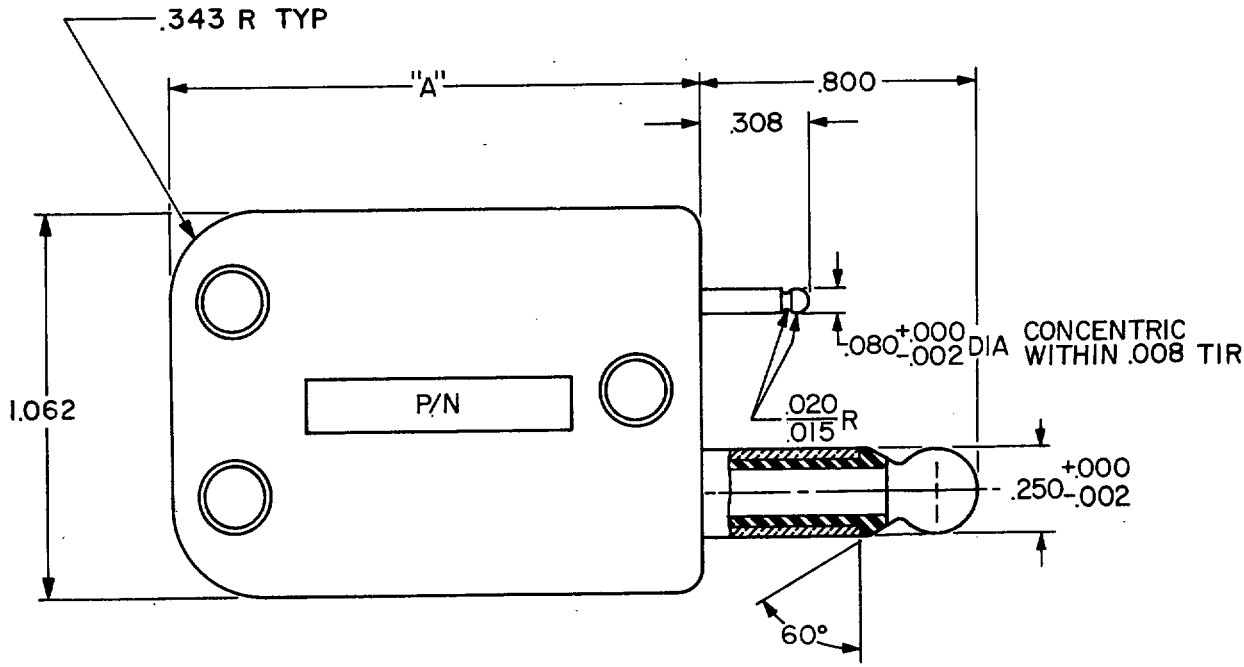


TABLE I	
DIMENSION A	PART NUMBER
1.5	M22442/5-1 (U-75/U)
2.0	M22442/5-2

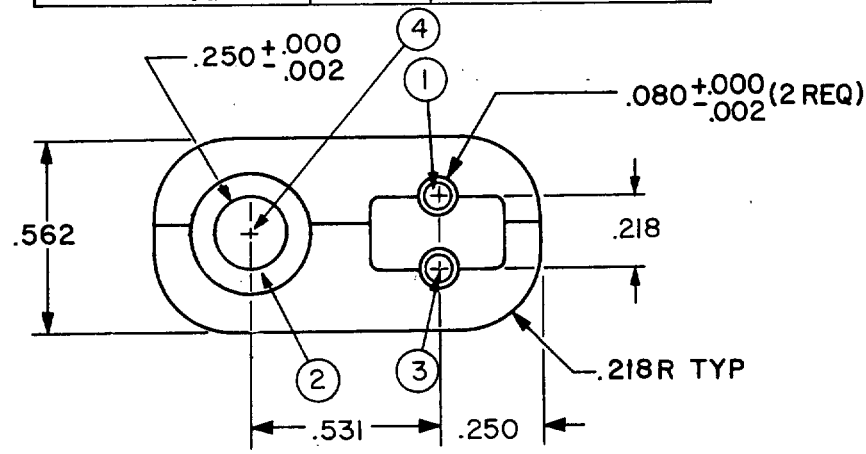


FIGURE 1. Connector (U-75/U)

FSC 5995

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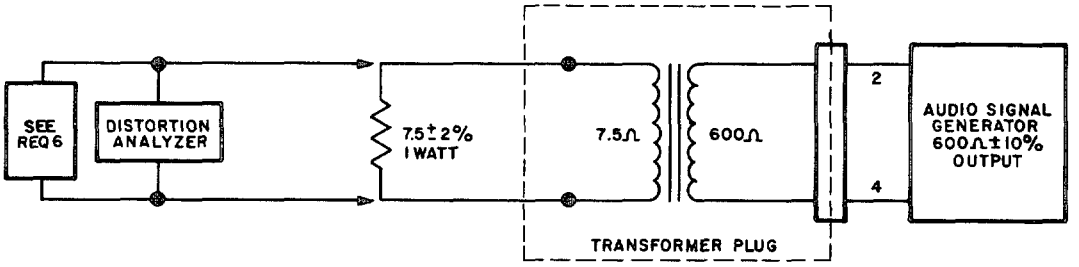


FIGURE 2. Test circuit.

The M22442/5-1 connector shall meet Requirements 1 thru 7. The M22442/5-2 connector shall meet Requirements 1 thru 11.

REQUIREMENTS:

1. M22442/5-1 (U-75/U) and M22442/5-2 shall be capable of mating with M22442/4-1 (U-61/U).

2. Sampling and testing procedures shall be in accordance with MIL-P-642.

3. The connector shall pass the following tests when tested in accordance with MIL-P-642.

   a. Visual and mechanical
   b. Dielectric strength
   c. Contact resistance
   d. Insertion and withdrawal force
   e. Spring life
   f. Salt spray
   g. Insulation resistance
   h. Moisture resistance

4. The insertion force shall be 8 lbs maximum. The withdrawal force shall be 6 $\pm$ 2 lbs.

5. After the spring life test the withdrawal force shall be greater than 60 percent of the withdrawal force measured prior to test.

6. The connector insulating materials shall conform to MIL-M-14 type SDG or MIL-M-21347 Type II.

7. Metal parts shall meet all the requirements and tests of MIL-P-642.

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8. The M22442/5-2 connector shall contain a transformer of the isolating impedance matching type capable of transferring audio frequency energy from a 600 ohm source (radio receiver output) to a 7.5 ohm load (two 15 ohm impedance earphones connected in parallel). The transformer shall provide a maximum power transfer of one-half watt with a power insertion loss not to exceed 2.5 dB with no dc flowing in either primary or secondary windings.

   a. The frequency response characteristics of the transformer shall be smooth and free from abnormal irregularities over the frequency range of 300 to 6000 Hz. The output response over this frequency range shall be within ±3 dB of the 1000 Hz output when tested with the test circuit shown.

   b. The distortion introduced by the transformer shall not exceed 5 percent for frequencies of 1000 to 6000 Hz. The distortion may increase for frequencies below 1000 Hz but shall not exceed 15 percent at 300 Hz when tested with the test circuit shown.

   c. The impedance of the primary and secondary windings over the frequency range of 300 to 6000 Hz shall not deviate from the 1000 Hz value by more than 15 percent when tested with the test circuit shown.

   d. The power loss shall not be greater than 2.5 dB at 1000 Hz.

9. An electronic voltmeter with an input impedance greater than 1 megohm shall be used.

10. The transformer shall be connected to contacts 2 and 4.

11. The cable assembly shall be tested in accordance with the tests listed in Table I.

NOTES:

1. Dimensions in inches. Unless otherwise specified, tolerances: linear ±.015.

Preparing Activity
Navy - AS
Project No. 5995-N110-20

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TABLE I

INSPECTION	REQUIREMENTS PARA.	TEST PARA.	GROUP I	GROUP II	GROUP III
			QUALIFICATION TESTS	QUALITY CONFORMANCE TEST 100%	SAMPLE TESTING PER 4.5.1
GROUP A (ALL SAMPLE UNITS)					
VISUAL & MECHANICAL INSPECTION	3.1, 3.4, 3.6	4.6.2	X	X	
SOLDERING	3.4.24	4.6.2.1	X	X	
GROUP D (3 SAMPLES) 1/					
IMPEDANCE	3.5.1	4.6.3	X		X
FREQUENCY RESPONSE	3.5.20	4.6.21	X	X	
DISTORTION	3.5.22	4.6.21	X		X
POWER LOSS			X		X
GROUP E (4 SAMPLES)					
TEMPERATURE-ALTITUDE	3.6.1	4.7	X		
SALT SPRAY	3.6.1	4.7	X		
THERMAL SHOCK	3.5.9	4.6.11	X		
VIBRATION	3.6.1	4.7	X		X

1/ TESTED IN ACCORDANCE WITH MIL-C-22442/1-3.

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